LAB EXPERIMENT RECORD

Set 25: Website Traffic Analysis

Department	AIDS
Course Code / Course Name	DSA0602/ Data Handling and Visualization
Experiment	Data Visualization using R
Name	Swetha S
Register Number	192424049
Date	20/10/2026
Day	Tuesday

AIM

To analyze website traffic data using R and Tableau by creating a line chart, a top-N click-through-rate bar chart, a stacked area chart for user interactions, and an interactive dashboard showing traffic sources and page views.

4.

- Prepare a dataset containing Traffic Source, Location, and Page Views.
- Open Tableau and connect to the dataset.
- Create a map by assigning geographic fields such as Country or City to the map.
- Drag Page Views to Color or Size to show traffic volume by location.
- Create a bar chart by dragging Traffic Source to Columns and Page Views to Rows.
- Create a new dashboard and add both the map and bar chart.
- Enable filter actions so selecting a location or traffic source updates the other chart.
- Add a suitable title, labels, and legends.

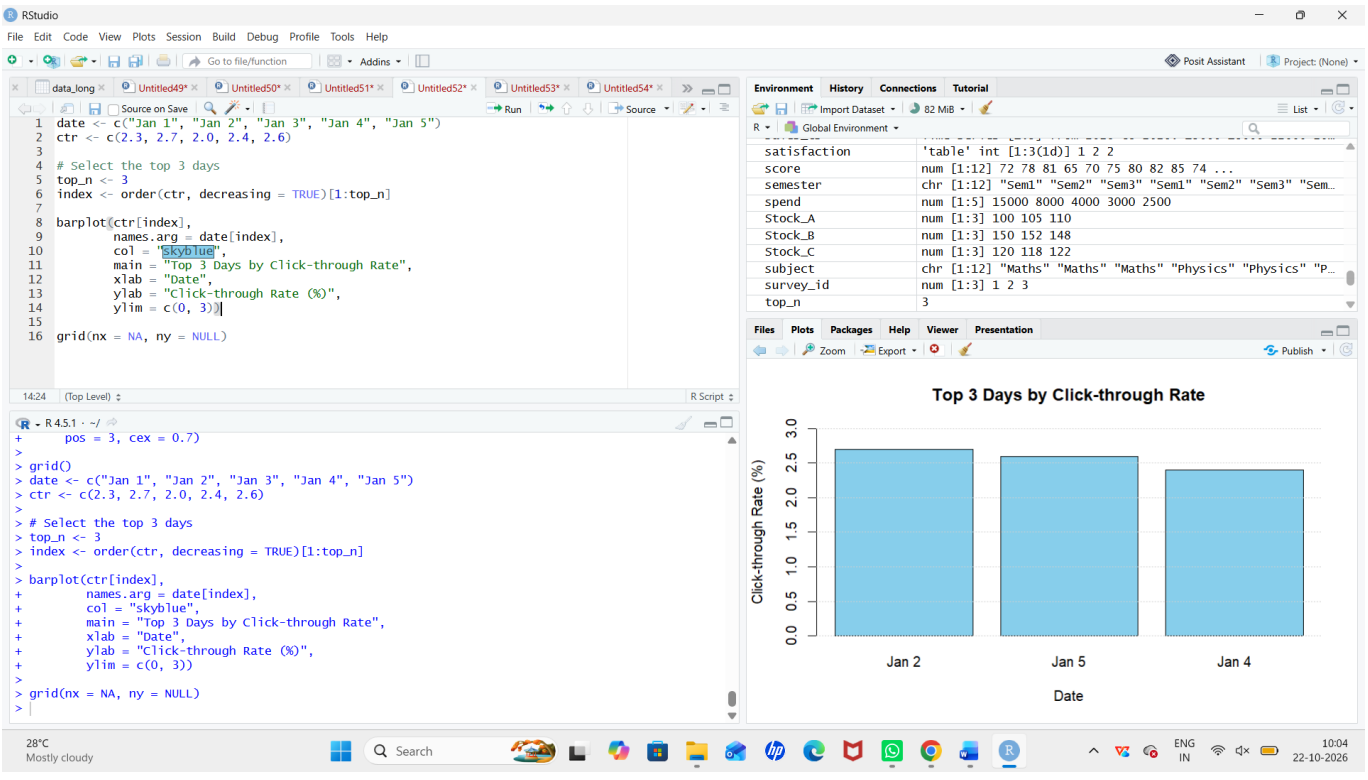
2

```
date <- c("Jan 1", "Jan 2", "Jan 3", "Jan 4", "Jan 5")
ctr <- c(2.3, 2.7, 2.0, 2.4, 2.6)

# Select the top 3 days
top_n <- 3
index <- order(ctr, decreasing = TRUE)[1:top_n]

barplot(ctr[index],
        names.arg = date[index],
        col = "skyblue",
        main = "Top 3 Days by Click-through Rate",
        xlab = "Date",
        ylab = "Click-through Rate (%)",
        ylim = c(0, 3))

grid(nx = NA, ny = NULL)
```



3.

```
date <- c("Jan 1", "Jan 2", "Jan 3", "Jan 4", "Jan 5")

likes <- c(100, 120, 90, 130, 150)
shares <- c(40, 50, 35, 55, 60)
comments <- c(25, 30, 20, 35, 40)

interaction <- rbind(likes, shares, comments)

colnames(interaction) <- date

matplot(1:5, t(interaction),
        type = "n",
        xaxt = "n",
        xlab = "Date",
        ylab = "Number of Interactions",
        main = "Website User Interactions")

grid()

stackpoly <- function() {
  polygon(c(1:5, 5:1),
          c(interaction[1, ], rep(0, 5)),
          col = "skyblue", border = NA)

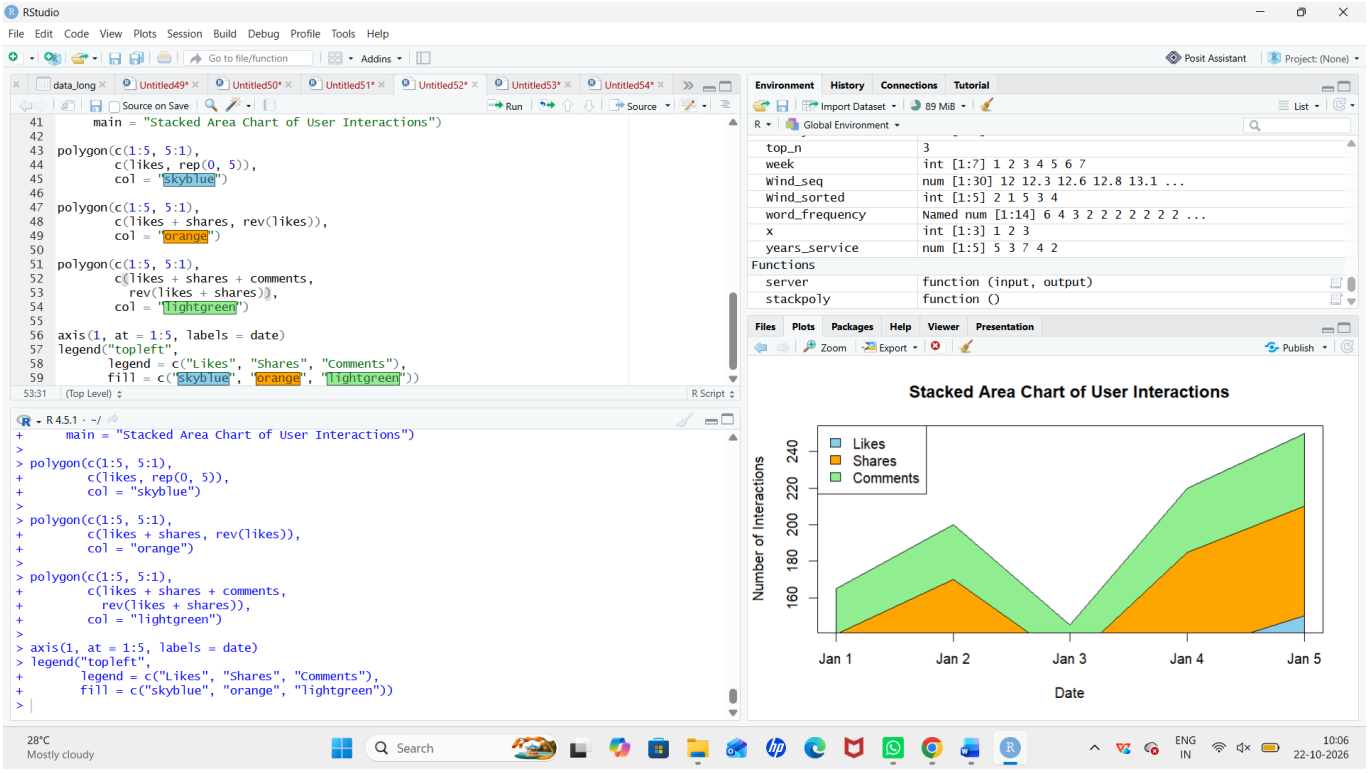
  polygon(c(1:5, 5:1),
          c(colSums(interaction[1:2, ]),
            rev(interaction[1, ])),
          col = "orange", border = NA)

  polygon(c(1:5, 5:1),
          c(colSums(interaction),
            rev(colSums(interaction[1:2, ]))),
          col = "lightgreen", border = NA)
}

# Draw the stacked areas from bottom to top
plot(1:5, colSums(interaction),
     type = "n", xaxt = "n",
     xlab = "Date",
     ylab = "Number of Interactions",
     main = "Stacked Area Chart of User Interactions")

polygon(c(1:5, 5:1),
        c(likes, rep(0, 5)),
        col = "skyblue")

polygon(c(1:5, 5:1),
        c(likes + shares, rev(likes)),
        col = "orange")
```



1.

- ☐ Prepare the dataset containing Date, Page Views, and Click-through Rate.
- ☐ Open Tableau and connect to the dataset.
- ☐ Drag Date to Columns.
- ☐ Drag Page Views to Rows.
- ☐ Select the Line Chart visualization.
- ☐ Add the title “Website Page Views Over Time” and label both axes

Procedure

1. Prepare the website traffic dataset containing dates, page views, and click-through rates.

2. Import and define the dataset in R.

3. Create a Tableau line chart to visualize page views over time.

4. Generate an R bar chart to identify the top days with the highest click-through rates.

5. Create a stacked area chart to compare likes, shares, and comments using sample interaction data.

6. Import the required traffic data into Tableau.

7. Create an interactive map showing traffic sources by geographic location and a bar chart displaying page views by source.

8. Combine both visualizations into a Tableau dashboard and add suitable filters, titles, and labels.

9. Analyze the charts to identify website traffic trends and user engagement patterns.

Result

Website traffic data was analyzed using R and Tableau. The line chart displayed page-view trends, the bar chart identified the days with the highest click-through rates, and the stacked area chart illustrated sample user interactions. The Tableau dashboard was designed to compare geographic traffic sources and page views. Additional interaction and location data are required for the corresponding visualizations.